PDF Compressor

A Web-Based Application for Reducing PDF File Size

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Project Type	Web Application (Spring Boot)
Category	File Utility / Productivity Tool

1. What the Website Is

The PDF Compressor is a browser-based tool that lets a user upload a PDF file and receive back a smaller version of the same file, without leaving the page. The user opens the site, drags a PDF into the upload area (or selects one manually), chooses a compression level - Low, Medium, or High - and clicks a single button. Within seconds, the page displays the original file size, the new compressed size, and the percentage of space saved, along with a download link for the compressed file.

The application is built with Java and Spring Boot on the backend, Apache PDFBox for the actual compression logic, and a lightweight HTML/CSS/JavaScript interface on the front end. No installation is required by the end user beyond opening a web page; no account, sign-up, or payment is involved.

2. The Problem

Large PDF files are a common everyday annoyance. Scanned documents, reports full of photos, and exported presentations often end up several times larger than they need to be, mainly because of high-resolution embedded images. This causes several practical issues:

- Email attachment limits are exceeded, forcing users to find workarounds to send a single document.
- Uploading documents to portals, job applications, or government forms fails when file-size limits are enforced.
- Storage and bandwidth are wasted, especially for users on limited data plans or slow connections.
- Existing 'free' online PDF compressors are frequently ad-heavy, add watermarks, upload files to unknown third-party servers, or place daily limits behind a paywall.

The core problem, therefore, is: how can a user shrink a PDF quickly, privately, and for free, without compromising the readability of the document?

3. My Solution

I built a self-hosted, single-purpose web application that solves this directly:

- **Smart image recompression, not blind file shrinking.** The application scans every page of the uploaded PDF, finds embedded raster images, and re-encodes them as JPEG at a quality level tied to the user's chosen compression setting. Text, fonts, and vector graphics are left completely untouched, so the readable content of the document is never degraded.
- **Three compression levels.** Low preserves near-original image quality for light savings; Medium balances quality and size for everyday use; High is for maximum size reduction when quality can be sacrificed, such as for scanned paperwork.
- **Privacy by design.** Uploaded files are processed entirely in memory on the server and are automatically discarded after a short time - nothing is written permanently to disk, and no file is shared with any third party.
- **Instant feedback.** The interface immediately shows the original size, the compressed size, and the percentage saved, so the benefit of compressing is visible before the user even downloads the result.
- **No cost, no limits, no watermark.** Because it runs as a self-contained Spring Boot application, it can be hosted anywhere and used without the restrictions of typical free online compressor tools.

The result is a tool that directly targets the root cause of oversized PDFs - unoptimized embedded images - while keeping the process fast, private, and free to use.

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